

Избранное

Перестаньте строить графики: когда GraphRAG действительно превзойдет vector RAG

Dattaraj Rao, Persistent Systems

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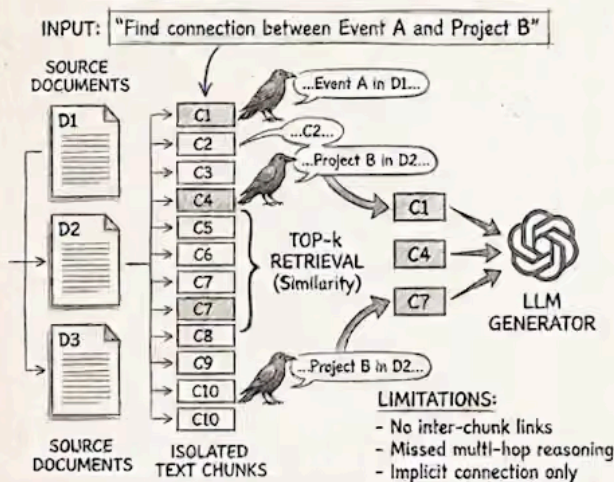


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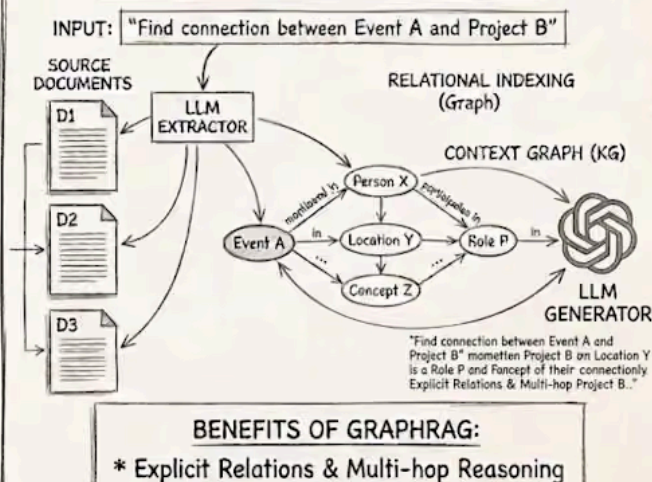


RAG PERFORMANCE COMPARISON

TRADITIONAL RAG: TEXT CHUNKS



CONTEXT GRAPH RAG (GraphRAG)



Предоставлено автором

Если вы создавали что-либо с помощью [поколения с расширенным поиском](#) (RAG) за последние два года, вы столкнулись с его основным разочарованием: вы разбиваете свои документы на фрагменты, вставляете их, извлекаете несколько лучших, которые выглядят похожими на вопрос, и передаете их модели. Для вопроса «Какова была наша политика возврата средств в третьем квартале?» этот вариант отлично подходит. Для вопроса «Какие темы повторяются в жалобах клиентов за последние два года?» он не подходит, потому что ни один фрагмент *не содержит* ответа.

Модное решение — **GraphRAG**: вместо того чтобы давать модели отдельные фрагменты текста, вы сначала создаете граф знаний об объектах и связях в вашем корпусе, а затем используете эту структуру в качестве контекста. Звучит заманчиво. Но заманчивые предложения заслуживают тщательного изучения, поэтому я изучила доказательства — оригинальную [статью Microsoft](#) и четыре независимых [сравнительных исследования](#), — чтобы ответить на простой вопрос: действительно ли ответы становятся лучше, когда вы заменяете

Краткая версия: да, в значительной степени — но только для правильных вопросов и не бесплатно. Позвольте мне показать вам доказательства.

Почему текстовые фрагменты упираются в стену

Стандартный [векторный метод RAG](#) извлекает k фрагментов, наиболее похожих на ваш запрос. У этой конструкции есть три структурных недостатка:



- **Она не может связать все воедино.** Когда для ответа требуется объединить факты, содержащиеся в разных фрагментах, через общую сущность, фрагменты, взятые по отдельности, никогда не покажут эту связь.
- **Он не учитывает глобальные вопросы.** «Каковы основные темы?» требует анализа *всего* корпуса, но поиск по сходству возвращает лишь несколько фрагментов, которые лишь внешне напоминают вопрос.
- **Контекст разрывается на границах чанков.** Отношения и иерархия, от которых зависит сложное мышление, — это именно то, что отбрасывается при чанкировании.

[Microsoft Research](#) framed this crisply when they introduced GraphRAG: Baseline RAG “struggles to connect the dots” and performs poorly when asked to “holistically understand summarized semantic concepts over large data collections.”

What a context graph changes

GraphRAG attacks the problem before any question is asked. During indexing, a large language model (LLM) reads every chunk and extracts entities, relationships, and claims, assembling them into a weighted knowledge graph. It then runs community detection (the [Leiden algorithm](#)) to cluster the graph into a hierarchy of related topics, and pre-writes a natural-language summary for each community.

At query time, those summaries do the heavy lifting. Each relevant community drafts a partial answer (the “map” step), the partials are ranked and merged (the “reduce” step), and the model synthesizes a final response grounded in structure rather than in a few cherry-picked snippets. Variants like [HippoRAG](#) take a different route, using the graph plus a Personalized PageRank walk to *find* the right passages — but the core idea is the same: Let relationships, not just cosine similarity, decide what context the model sees.

The evidence: Four studies, one pattern

1. Global sense making: The headline win

Microsoft pitted GraphRAG head-to-head against naïve RAG on global, “make sense of the whole corpus” questions over million-token datasets, with an LLM acting as judge across three axes: Comprehensiveness, diversity, and empowerment.

GraphRAG won 72 to 83% of comprehensiveness comparisons and 62 to 82% of diversity comparisons against [vector RAG](#). Its highest-level summaries used up to 97% fewer tokens than processing the source text directly.

That is not a rounding-error improvement. On exactly the kind of question that breaks text-chunk RAG, the graph wins two out of three times or better.

2. Multi-hop retrieval: The graph finds what chunks miss

The second piece of evidence is about retrieval quality: Does the right supporting passage even make it into the top results? On the standard multi-hop QA benchmarks (MuSiQue, HotpotQA, 2WikiMultiHopQA), graph-guided retrieval lifts Recall@5 dramatically:

- Average Recall@5 climbs from **73.4% (naïve RAG)** to **87.8% (graph-guided)**, a **+19.6 point** gain.
- The biggest jumps come on the hardest, cross-document sets: **+31 points on MuSiQue** and **+28 points on 2Wiki**.
- HippoRAG reports up to a **20% accuracy improvement** on multi-hop QA, at **10–20× lower cost** and **6–13× faster** than iterative retrieval methods.

3. The controlled head-to-head - where it gets honest

Here is where the story gains nuance. A [2025 study](#) from Michigan State and Meta ran RAG against four GraphRAG families under one unified protocol — identical chunking, embeddings, and generation — and found no single winner. The two approaches are complementary:



- On single-hop, factual lookup (natural questions), plain RAG edged ahead (F1 64.8 vs. 63.0 for the best graph method).
- On multi-hop reasoning (MultiHop-RAG), graph-guided retrieval pulled in front (70.3 vs. 67.0 overall accuracy).

The lesson: A context graph is not a universal upgrade. It is a specialized one that pays off precisely when questions demand reasoning across pieces.

4. When to use graphs: The task-type verdict

The most recent benchmark, [GraphRAG-Bench](#) (ICLR 2026), set out to answer “In which scenarios do graph structures provide measurable benefits?” Its accuracy-by-task numbers map the boundary cleanly:

- **Simple fact retrieval:** Text chunks 60.9 vs. graph 60.1 — effectively a tie. The graph’s structure is overhead the query doesn’t need.
- **Complex reasoning:** Graph 53.4 vs. chunks 42.9 — a **+10 point** graph win.
- **Contextual summarization:** Graph 64.4 vs. chunks 51.3 — a **+13 point** graph win.

The scorecard



Study	Task — metric	Text chunks	Context graph
Microsoft GraphRAG (2024)	Global sensemaking — comprehensiveness	baseline	72–83% win
Microsoft GraphRAG (2024)	Global sensemaking — diversity	baseline	62–82% win
HippoRAG (NeurIPS 2024)	Multi-hop QA — accuracy	baseline	up to +20%
Multi-hop benchmarks	Recall@5 — average	73.4%	87.8%
Han et al. (2025)	Single-hop QA (NQ) — F1	64.8	63.0
Han et al. (2025)	Multi-hop overall — accuracy	67.0	70.3
GraphRAG-Bench (2026)	Complex reasoning — accuracy	42.9	53.4
GraphRAG-Bench (2026)	Simple fact retrieval — accuracy	60.9	60.1

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Read top to bottom, the pattern is unmistakable: The graph’s advantage grows with the reasoning depth of the question, while text chunks hold their ground on isolated facts.

The catch: Cost and the LLM-judge problem

Two caveats keep this from being a slam dunk, and ignoring them is how teams end up disappointed.

Building the graph is expensive. Having an LLM extract entities and relationships from an entire corpus isn’t cheap. One analysis put index construction at roughly \$48 against GPT-4o for a moderate corpus, far above a vanilla vector index. (Microsoft’s own follow-up, LazyGraphRAG, defers extraction to query time and cuts that to around 0.1% of the cost – a tacit admission that the original budget is impractical for many deployments.)

Many of the wins are judged by another LLM — and [LLM judges](#) are biased. An

length bias, and **trial bias** (identical comparisons disagree across runs). After correction, one popular method's reported 66.7% win rate fell to about **39%** — below the 50% break-even line.

The takeaway is not “the research is wrong.” It is that the large gains — the +20% multi-hop accuracy, the +15-to-30-point recall jumps — are robust, while narrow comprehensiveness margins deserve a skeptical second look with reference-based metrics.

So when should you reach for a context graph?

Strip away the hype and the decision is refreshingly practical.

Use a context graph when: Your questions are multi-hop, global, or sensemaking in nature; you need comprehensive, multi-perspective answers; and your corpus is richly interconnected (research libraries, case files, incident histories, knowledge bases).

Stick with text chunks when: Your queries are mostly single-fact lookups; your corpus is small or flat; and indexing cost, latency, and operational simplicity outweigh a marginal quality bump.

Best of all, go hybrid: The systematic studies converge on the same recommendation: route each query to the right method, or fuse evidence from both. Combining graph and chunk retrieval consistently beats either one alone. You don't have to choose a religion; you have to build a router.

The bottom line

A context graph is not magic, and it is not snake oil. It is a targeted instrument. Hand it a question that requires connecting scattered facts or synthesizing a whole corpus, and it will outperform text chunks decisively. Hand it “what's the phone number on page 3,” and you've paid for indexing you didn't need.

The teams that win with GraphRAG in 2026 won't be the ones who graph everything. They'll be the ones who know which questions deserve a graph — and build pipelines smart enough to tell the difference.

Dattaraj Rao is an R&D architect at Persistent Systems



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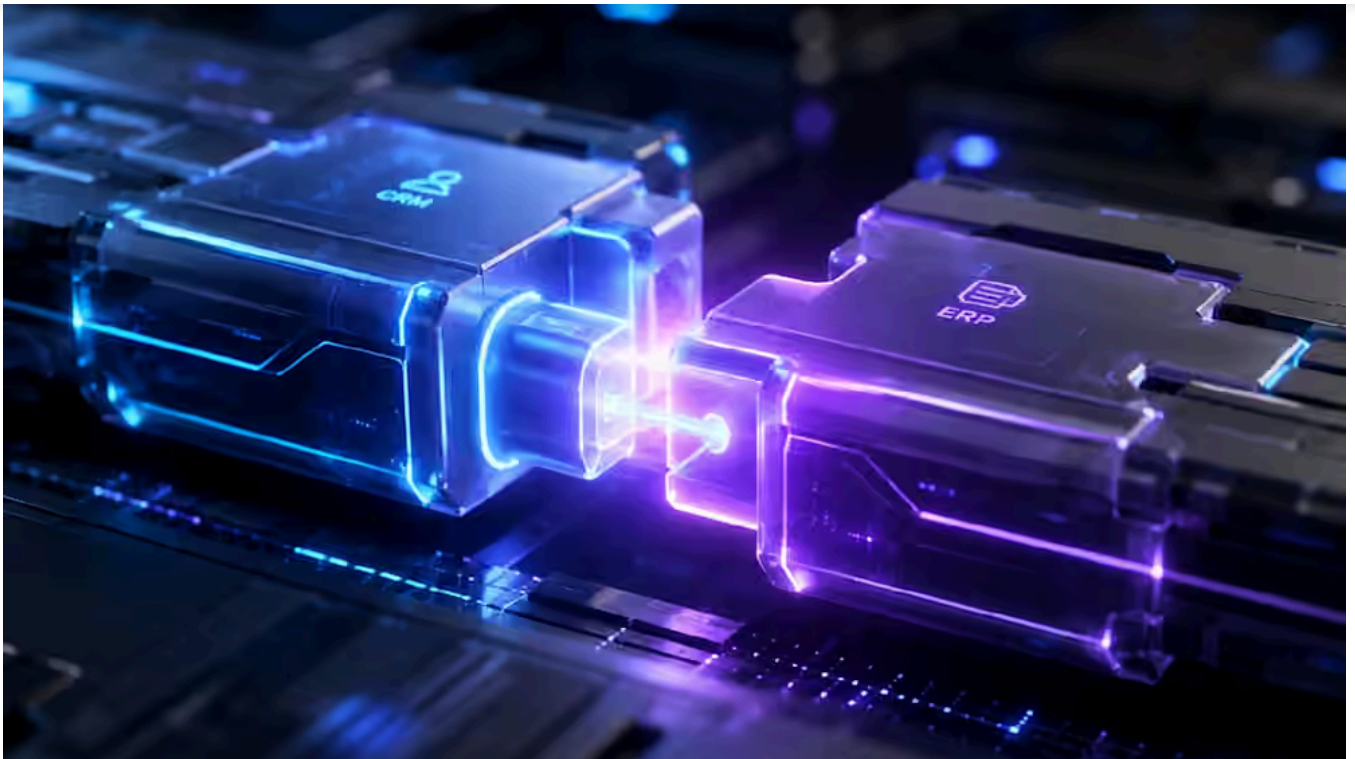
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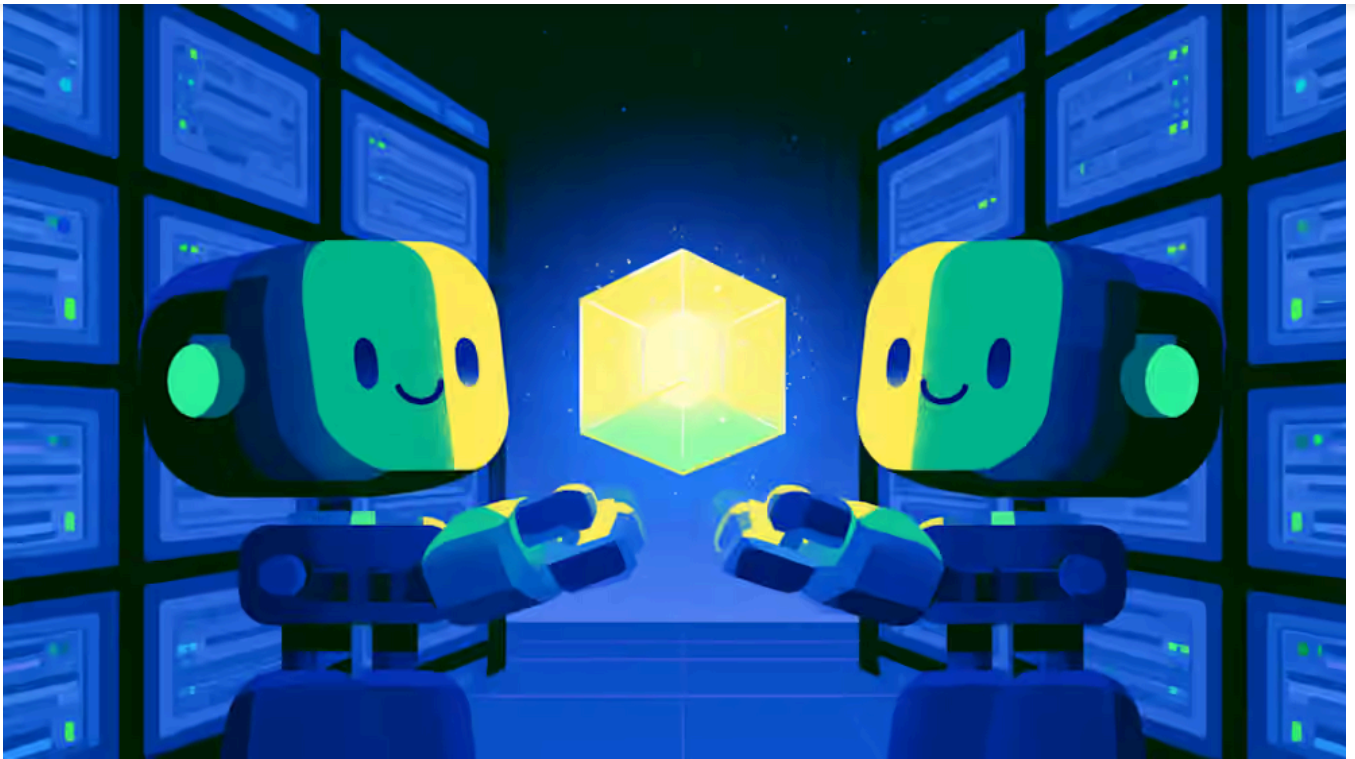
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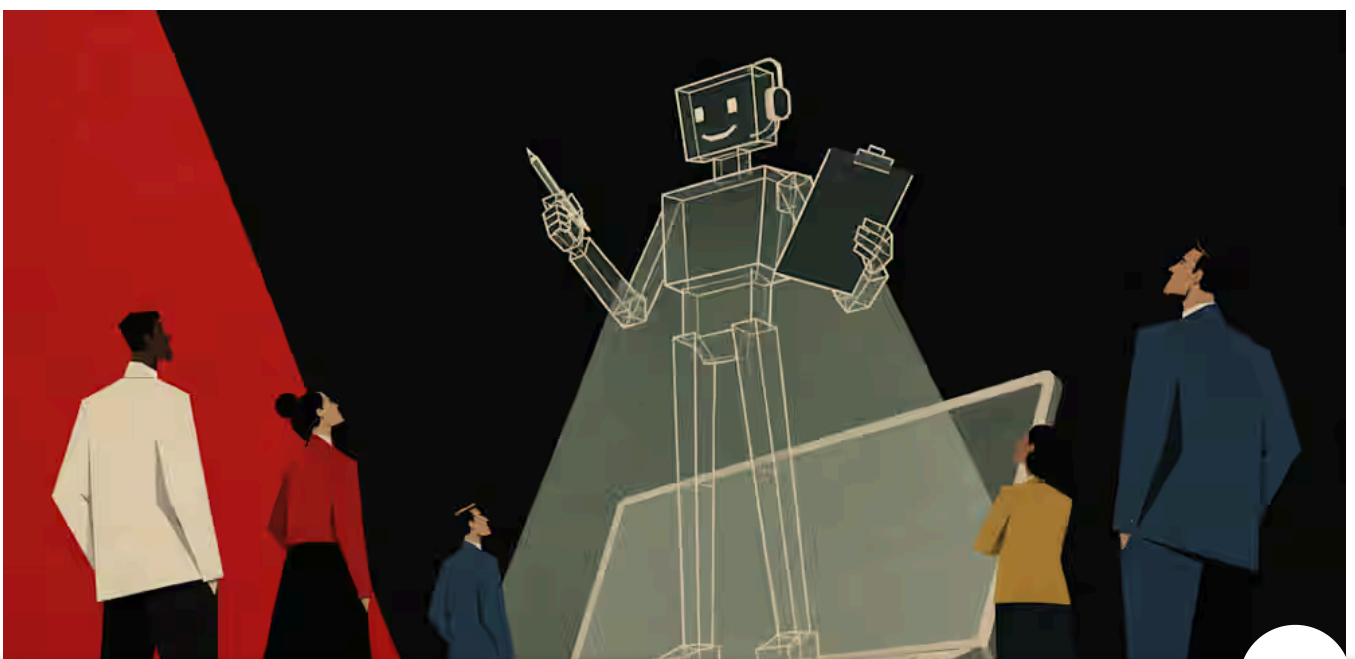


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